How to Become a Data Architect (With Essential Skills)

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Data architects are IT experts who design and manage data systems, develop and set policies for data storage and access, and incorporate new data technologies into existing infrastructure. They typically work in teams that include other data professionals like data miners, data scientists, and data analysts. Understanding how to become a data architect can help increase your chances of receiving an invitation to interview for a position. In this article, we discuss the necessary steps to become a data architect, highlight the essential skills you need, outline professional certifications for data architects, and identify their primary duties.

Key takeaways:

- Data architects usually require a background in computer science or a related field, practical experience (including in information technology), technical and business skills, and professional certifications.
- There are various skills of use to a data architect, including programming, data modelling, applied mathematics, experience with machine and natural learning, communication, industry knowledge, data visualization, and problem-solving.
- Obtaining related certification can be of benefit in the role of a data architect, and can include: Certified Data Management Professional (CDMP), IBM Certified Data Architect - Big Data, Salesforce Certified Data Architecture and Management Designer, Simplilearn's Big Data Architect Masters Program, and the TOGAF 9 Certification Program.

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How to become a data architect

If you're interested in this field, here's an overview of how to become a data architect:

1. Obtain a degree in computer science or a related field

Completing a university program in computer science or computer engineering is fundamental to a data architecture career. These programs involve courses on data management, big data development, technology architecture, system analysis, and computer programming. You can also choose electives in courses related to mathematics, statistics, or probability. These courses can provide you with essential background knowledge in math and statistics.

2. Gain experience

Experience is very important when starting a data architecture career. You can develop your experience in addition to obtaining a university degree to improve your chances of getting hired. Build experience by applying for relevant internships, taking university courses with practical requirements, and using real-life data. Consider developing a portfolio to help you track your experience and demonstrate it to the hiring manager.

3. Develop technical and business skills

Technical skills, like data mining and analytical problem-solving, are essential for data architects, so developing them is essential. There are various resources that can help you develop these skills. You can also consult with mentors in the industry and have them share their knowledge and insights with you. You can use these skills to work on challenges as this helps you grow professionally.

4. Gain work experience in the information technology field

To get into data architecture, you may first need experience in a related field, like IT. You can apply for entry-level jobs in programming or database administration to help build your resume. Working in these positions for about three to five years can help you gain the necessary experience for data architecture. You can use this period to develop your skills in development management, warehousing, modelling, and database design.

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5. Get hired as a data architect

Applying for positions in data architecture is the next step after gaining related experience. You can search for jobs in finance firms, healthcare firms, large educational entities, insurance companies, and businesses. These companies gather and process large amounts of client data, and they often require the services of a data architect. Other companies with opportunities for data architecture include software corporations and computer manufacturing companies.

6. Obtain professional certifications

Professional certifications can provide you with more opportunities and higher pay. There are several professional certifications that can help you expand your expertise in data architecture. You can determine the certifications that are best for you by examining recent job descriptions and consulting mentors. The Certified Data Management Professional (CDMP) is the most well-known professional certification in this field. The certificate requires that candidates have two years of work experience and a bachelor's degree before taking the certification exam.

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7. Obtain a postgraduate degree

A master's degree in a related program can increase your job opportunities and rate of pay, similar to professional certifications. For example, employers often prefer those with a master's degree for senior positions in data architecture. You can consider an advanced degree in data architecture or a related field.

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Essential skills for a data architect

Here is a list of skills to consider if you are interested in becoming a data architect:

Programming skills

Programming skills are essential for data architects, and some languages carry particular importance. For example, Structured Query Language (SQL) is an essential and fundamental language for data architects. SQL helps to access, manipulate, manage, and query data from relational database models. Another vital programming language is Python. Python can help handle big data or connect to database systems. It's a multi-purpose language that can develop several applications for data analysis.

Data modelling

Data modelling is another essential skill for a data architect. It involves representing the manner of data flow through simple diagrams of a complex software system. Data models assist stakeholders in detecting errors and resolving them before writing a programming code.

Likewise, data modelling demonstrates a data architect's understanding of a data model and its design principles. Several online and offline resources can help you learn and master data modelling.

Applied mathematics, statistics, and analytical skills

A data architect's job includes analytical reasoning, interpretation, and computation. These require a background in mathematics and statistics. Mathematics and statistics are also essential for other data professions, so you can apply yourself in other data fields without learning completely new skills.

Machine learning and natural learning processing

Knowledge of machine learning, natural learning processing, and pattern recognition is essential for data architects. This is because understanding the interaction between computer and human language is vital for data architects to solve data problems. Knowledge of these tools is also necessary as they help data architects to use clusters in text mining and data handling.

Communication skills

Senior data architects are typically in charge of a team of other data professionals, like data engineers or database administration. A data architect requires efficient communication skills to communicate plans and instructions to team members successfully. Effective communication skills can help you communicate solutions to colleagues with a non-technical background.

Knowledge of the industry

Data architects need an in-depth knowledge of their industry. This includes the trends and best practices regarding data collection, storage, and analysis. Typically, a certification demonstrates sufficient industry knowledge, but it's a data architect's duty to stay up-to-date regarding new industry technology or practices.

Data visualization and presentation

Data visualization involves presenting data in graphical styles to express insights from data. These skills are fundamental for several data professionals, including data architects. They assist the data architect in representing data in more understandable formats. There are several tools and resources that can increase your proficiency with data visualizations.

Problem-solving skills

Problem-solving is a necessary soft skill for data architects. These professionals regularly encounter challenging situations that require quick and reliable judgement to resolve. You can develop your problem-solving skills in data architecture by researching or asking questions from teammates.

Examples of certifications for data architects

Here are some of the essential professional certifications for you to consider if you are interested in pursuing a career in data architecture:

Certified Data Management Professional (CDMP)

The Data Management Association International (DAMA) administers this professional certification. CDMP is a standard certification because it doesn't focus on a specific vendor or platform. This lack of specificity makes it essential for general database professionals. The certification comes in four levels, namely, associate, practitioner, master, and fellow. You require evidence of education, experience, and passing a professional exam to receive the certification.

IBM Certified Data Architect - Big Data

This IBM professional certification requires candidates to have numerous prerequisite skills. These include understanding cluster management, data lineage, data replication, and LDAP security. The exam focuses on BigSQL, BigInsights, Cloudant, and Hadoop.

Salesforce Certified Data Architecture and Management Designer

This professional certification is for candidates with experience working with the Salesforce platform. The professional certification exam typically tests knowledge of large data volume risks and mitigation approaches. The exam also assesses the candidate's understanding of design trade-offs, LDV considerations, and best practices in an LDV environment.

Simplilearn's Big Data Architect Masters Program

This certification, offered by Simplilearn, helps to expand knowledge of several vital tools. These tools include data modelling, database interfaces, Scala, replication SparkSQL, Hadoop clusters, and RDD. Candidates usually learn from instructors and by participating in lab experiences and projects for this certification.

TOGAF® 9 Certification Program

The TOGAF 9 certification occurs in two forms, foundation and certification. The certification form requires the completion of a professional exam. In contrast, the foundation aspect ensures that candidates have sufficient knowledge of basic TOGAF 9 concepts and core principles.

Related: What Is a TOGAF Certification? (With Benefits and Tips)

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Primary duties of data architects

Here are some responsibilities of a data architect:

- collaborating with IT teams to develop data strategies that meet and address industry requirements
- maintaining an organization's corporate repository of its data architecture procedures and artifacts
- creating data inventories to implement the data architecture
- developing and enforcing database development standards
- detecting and assessing current data management trends and technologies
- observing, refining, and reporting on the operation of data management systems
- building fluid data models for data structures
- developing, implementing, and enforcing measures to ensure data accessibility and accuracy
- incorporating technical functionality such as security, performance, scalability, or reliability
- developing, constructing, deploying, and documenting database architectures and applications like large relational databases
- incorporating new data structures into existing organizational structures

Now that we've discussed the necessary steps regarding how to become a data architect, and highlighted the essential skills you need, you'll be better prepared to decide if this is a career path for you.

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